A derivative singlet and its two-point spectrum in complex double-scaled SYK

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Local research note — 6 November 2026

Abstract

The first derivative singlet is made from the same couplings as the Hamiltonian. We reduce its complete, ensemble-averaged two-point spectral measure at infinite temperature to a conserved quadratic channel and the known q-Hermite probe kernel with crossing parameter q^2. The reduction follows from an adjacent-endpoint exchange identity for marked chord diagrams at every fixed moment order. It extends our previous two-moment calculation to fixed real times and fixes the limiting zero-frequency weight. The microscopic-to-bulk comparison can now begin from an explicit propagator rather than from a fitted frequency scale.

1 The propagation channel

Our model and state remain

$$H = \sum_{|I|=|J|=p} G_{IJ} c_I^\dagger c_J, \qquad \mathbb{E} G_{IJ} G_{KL} = w \delta_{IL} \delta_{JK}, \tag{1}$$

$$w = 4^p \frac{\mathcal{J}_s^2}{\lambda} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda = p^2/N, \quad \mathrm{tr} = 2^{-N} \operatorname{Tr}. \tag{2}$$

The full Hermitian Gaussian ensemble includes overlapping index sets; $\{c_i, c_j^\dagger\} = \delta_{ij}$, $\hbar = 1$, and $H = H_s$ generates string time. All limits below first take $N, p \to \infty$ at fixed $\lambda > 0$. Put

$$A = \sum_i [H, c_i]^\dagger [H, c_i], \quad O = A - \mathbb{E}A, \quad B = H^2 - \mathbb{E}H^2, \quad q = e^{-4\lambda}, \quad \sigma^2 = \mathcal{J}_s^2/\lambda. \tag{3}$$

Both ensemble means are functions of number Q. Thus B is conserved in every realization. With $C_O(t) = \mathbb{E} \operatorname{tr} O(t) O$, the result is

$$\boxed{C_O^{\mathrm{DS}}(t) = W_0 + W_{\mathrm{d}}\, G_{q^2}(t), \qquad G_{q^2}(0) = 1.} \tag{4}$$

The weights are

$$\boxed{W_0 = 4\mathcal{J}_s^4 \frac{(1-q)^2}{1+q}, \qquad W_{\mathrm{d}} = 4\mathcal{J}_s^4 \left(\frac{1}{\lambda} + \frac{4q}{1+q} \right).} \tag{5}$$

Their sum reproduces $V_0 = 4\mathcal{J}_s^4(\lambda^{-1} + 1 + q)$. The strictly dynamical limiting channel is obtained by subtracting cB, with $c = 2\lambda(1-q)/(1+q)$. Its two-point function is $W_{\mathrm{d}} G_{q^2}(t)$. The proof is in Appendices A–C.

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2 An explicit kernel for the bulk comparison

The kernel itself is established in the chord literature [1, 2]; our task is the reduction of the same-coupling O to it. Define $(z;q)_\infty = \prod_{k\geq 0}(1-zq^k)$ and

$$E(\theta) = \frac{2\sigma\cos\theta}{\sqrt{1-q}}, \qquad \rho_q(\theta) = \frac{(q, e^{2i\theta}, e^{-2i\theta}; q)_\infty}{2\pi}, \tag{6}$$

$$K_r(\theta,\phi) = \frac{(r^2;q)_\infty}{\prod_{\epsilon,\eta=\pm1}(re^{i(\epsilon\theta+\eta\phi)};q)_\infty}, \qquad r = q^2. \tag{7}$$

Then the normalized propagator and its spectral measure are

$$G_r(t) = \int_0^\pi \mathrm{d}\theta \int_0^\pi \mathrm{d}\phi\, \rho_q(\theta)\rho_q(\phi)K_r(\theta,\phi)e^{it[E(\theta)-E(\phi)]}, \tag{8}$$

$$\mathrm{d}\mu_r(\omega) = \int_0^\pi \mathrm{d}\theta \int_0^\pi \mathrm{d}\phi\, \rho_q(\theta)\rho_q(\phi)K_r(\theta,\phi)\delta\big(\omega - E(\theta) + E(\phi)\big)\mathrm{d}\omega. \tag{9}$$

The measure is positive, symmetric, normalized, and absolutely continuous. Its support lies in $|\omega| \leq 4\sigma/\sqrt{1-q}$. The full singlet measure is $W_0\delta_0 + W_{\mathrm{d}}\mu_{q^2}$. In the convention $C(t) = \int e^{it\omega}S(\omega)\mathrm{d}\omega/(2\pi)$, S is 2π times this measure.

Small positive λ. The standard probe parameter $r = q^\Delta$ gives $\Delta = 2$. Applying the known leading small-deformation probe asymptotic [3], with $-\log q = 4\lambda$ and its coupling $\mathcal{J} = \sigma\sqrt{-\log q} = 2\mathcal{J}_s$, yields, at fixed string time,

$$G_{q^2}(t) \;\longrightarrow\; \mathrm{sech}^4(2\mathcal{J}_s t). \tag{10}$$

This is a reuse of the probe result, after establishing (4). The normalized limiting density corresponding to (10) is

$$S_{\mathrm{norm}}(\omega) = \frac{\pi\omega(\omega^2+16\mathcal{J}_s^2)}{96\mathcal{J}_s^4\sinh(\pi\omega/(4\mathcal{J}_s))}, \qquad \int\frac{\mathrm{d}\omega}{2\pi}S_{\mathrm{norm}}(\omega) = 1. \tag{11}$$

The removable value at $\omega = 0$ is $2/(3\mathcal{J}_s)$. Also $W_0/V_0 \sim 8\lambda^3$: the centered strength becomes predominantly dynamical in this order of limits.

Physical use. This supplies a concrete singlet propagation channel for the Susskind static-patch programme [5]. The label $\Delta = 2$ specifies the chord kernel. To assign a de Sitter particle mass we must match this kernel, its time coordinate, and its operator insertion to the proposed bulk theory. Higher singlet correlators are the next interaction test.

A From the microscopic operator to marked diagrams

Let $\mathcal{D}(X) = QX - \sum_i c_i^\dagger X c_i$. It acts by body rank on normal-ordered number-conserving monomials. Thus

$$A = (2p - \mathcal{D})H^2 = \partial_h R_h(H, H)|_{h=0}\,, \tag{12}$$

where R_h weights each normal-ordering contraction from the first annihilator block to the second creator block by $1 + h$. This is the contraction construction of our preceding note, restated here to fix the insertion. It is not an independently randomized probe.

For $n = a + b$ even, expand $\mathbb{E}\,\mathrm{tr}\, OH^aOH^b$ into $L = n + 4$ Hamiltonian slots. The marked adjacent pairs are $(1, 2)$ and $(a + 3, a + 4)$. Gaussian centering deletes exactly the Wick pairings that join the two slots within either marked pair. Call the remaining set $\mathcal{P}_*$.

For $\pi \in \mathcal{P}_*$, let $T_{ij} = \frac{1}{2}$ if $i \le j$ and $-\frac{1}{2}$ otherwise. At a marked pair (u, v) replace T_{vu} by $T_{vu} - h$ (or $-g$). The exact mode polynomial and paired word are

$$F_\pi(z; h, g) = \det(I + ZT(h, g)P_\pi), \tag{13}$$

$$\mathrm{word}_\pi = w^{L/2}(-1)^{Lp/2}[z_1^p \cdots z_L^p hg]F_\pi(z; h, g)^N. \tag{14}$$

The block permutation of the annihilator sets gives the displayed sign. Grouping rows and columns by fermion mode gives the local minors and accounts for all repeated mode labels.

No singleton minor contains a mark, since internally contracted marked pairs have been removed. Divide each local minor by its singleton factors. For any fixed L, pair overlaps of independent uniform p-sets tend jointly to independent Poisson variables of mean λ. Triple membership has probability at most $\binom{L}{3}p^3/N^2 = o(1)$. As in the preceding note, the hypergeometric bound

$$\mathbb{E}e^{s|I\cap J|} \le \exp\left[\frac{p^2}{N - p + 1}(e^s - 1)\right], \qquad s \ge 0, \tag{15}$$

and Hölder control the finite collection of overlaps. It supplies uniform integrability of the complex marked weights on a fixed neighborhood of the origin. Triple events vanish by Cauchy–Schwarz; Cauchy's formula then passes both derivatives through the limit. Consequently each graph contributes

$$\sigma^{n+4} q^{\mathrm{cr}(\pi)}(\lambda\delta_\pi + \lambda^2\alpha_\pi\beta_\pi). \tag{16}$$

The factorial and singleton factors in (14) cancel $w^{L/2}$; the unmarked exponent is $-4\lambda\,\mathrm{cr}(\pi)$. This argument holds at every fixed order, with no assumed uniformity in L.

B The endpoint exchange identity

The two-by-two minors in (14) give particularly simple rules. At a mark, let $\epsilon = +1$ when its two incident chords do not cross each other, and $\epsilon = -1$ when they do. Then

$$\alpha = 2\epsilon_1, \qquad \beta = 2\epsilon_2, \qquad \delta = 4\,\mathbf{1}_{\text{the two marks are joined by two nested chords}}. \tag{17}$$

These statements involve crossings of the incident chords with each other, not their crossings with any background chords.

Let $m_k(q)$ denote the ordinary unit-variance Hamiltonian chord moment, and let $P_{ab}(q)$ be the unit-normalized independent-probe mixed moment with crossing weight q^2. Put

$$u = \frac{1-q}{1+q}, \qquad S_n = m_{n+4} - 2m_{n+2} + m_n. \tag{18}$$

Deleting the two nested marked chords in the δ term leaves a background diagram. Each background chord straddling the two marks crossed both deleted chords, hence contributes q^2. Therefore

$$\sum_{\pi \in \mathcal{P}_*} q^{\mathrm{cr}(\pi)} \delta_\pi = 4P_{ab}. \tag{19}$$

For the $\alpha\beta$ term, exchange the two adjacent endpoints at one mark. This toggles only the crossing between its incident chords; all background crossings remain unchanged. In a two-element exchange pair the unsigned weights are w, qw and the signed weights $w, -qw$. The ratio of signed to unsigned sum is u.

Unless both incident chords join the two marks, the two endpoint exchanges have a four-element orbit. Their crossing choices are independent, even when the marks share one chord. Its signed sum is u^2 times its unsigned sum. When both chords join the marks, the orbit instead has two elements: nested and crossed. Both have $\epsilon_1\epsilon_2 = 1$, so its signed and unsigned sums agree. The total unsigned weight of these exceptional diagrams is $(1+q)P_{ab}$. The unsigned sum over all allowed diagrams is S_n: subtracting the two internal adjacent contractions is precisely $(H^2 - 1)$ at both marks in the unit-variance chord theory. Hence

$$\sum_{\pi \in \mathcal{P}_*} q^{\mathrm{cr}(\pi)} \alpha_\pi \beta_\pi = 4u^2 S_n + \frac{16q}{1+q} P_{ab}. \tag{20}$$

Combining (16)–(20) proves, for every $a, b \geq 0$,

$$\boxed{\lim_{\mathrm{DS}} \mathbb{E}\,\mathrm{tr}\, OH^a OH^b = c^2 \sigma^{n+4} S_n + W_{\mathrm{d}} \sigma^n P_{ab}, \quad c = 2\lambda u.} \tag{21}$$

For odd n, both sides vanish by Gaussian parity. One marked pair likewise gives $\lim \mathbb{E}\,\mathrm{tr}\, OH^n = c\sigma^{n+2}(m_{n+2} - m_n)$.

C Conserved projection and convergence of the measure

The ensemble Hilbert-space inner product is $(X, Y) = \mathbb{E}\,\mathrm{tr}\,X^\dagger Y$. At finite N, p let $c_{N,p} = (O, B)/(B, B)$ and $O_\perp = O - c_{N,p}B$. The one-mark identity yields

$$(B, B) \longrightarrow \sigma^4(1+q), \qquad (O, B) \longrightarrow 2\lambda\sigma^4(1-q), \tag{22}$$

$$c_{N,p} \longrightarrow c, \qquad \|c_{N,p}B\|^2 \longrightarrow W_0. \tag{23}$$

Here $\mathbb{E}H^2 \to \sigma^2 I$ in ensemble L^2: its trace and squared trace-norm follow from one and two prescribed internal adjacent Wick pairs. Each has limiting weight one. This justifies replacing its insertions by σ^2 in the mixed expressions by Cauchy–Schwarz and the bounded higher mixed moments. The centered B therefore becomes the second monic q-Hermite channel. The projection onto H itself vanishes by Gaussian parity; this is an ensemble statement.

For a direct spectral argument, define the positive finite measure

$$\nu_{N,p}(\mathrm{d}x, \mathrm{d}y) = \mathbb{E}\,\mathrm{tr}\big(O\,\Pi_H(\mathrm{d}x)\,O\,\Pi_H(\mathrm{d}y)\big). \tag{24}$$

Its mixed moments are (21). They agree with the positive candidate

$$\nu(\mathrm{d}x, \mathrm{d}y) = c^2(x^2-\sigma^2)^2\rho_E(x)\mathrm{d}x\,\delta_x(\mathrm{d}y) + W_{\mathrm{d}}\rho_E(x)\rho_E(y)K_{q^2}(x,y)\mathrm{d}x\mathrm{d}y, \tag{25}$$

where $\rho_E(x)\mathrm{d}x = \rho_q(\theta)\mathrm{d}\theta$ denotes the pushforward measure and $K(x, y)$ uses the corresponding angles. The known q-Mehler identity gives the second term's mixed moments [1]. Its support is the compact square with $|x|, |y| \le E_{\max} = 2\sigma/\sqrt{1-q}$.

Second energy moments give tightness of $\nu_{N,p}$; higher even moments give uniform integrability of each fixed monomial. Every weak subsequence limit therefore has these moments. Its marginals must be supported on $[-E_{\max}, E_{\max}]$, by arbitrarily high even moment bounds. Polynomials are dense on the compact square, so this candidate is unique. Thus $\nu_{N,p} \Rightarrow \nu$, and testing the bounded continuous function $e^{it(x-y)}$ proves (4) for each fixed t. This avoids exchanging an unbounded finite-N exponential series with the double-scaling limit.

The second term of (25) has an integrable density and no diagonal atom. Hence $C_O^{\mathrm{DS}}(t) \to W_0$ as $|t| \to \infty$. The finite-system zero-frequency weight is at least $(O, B)^2/(B, B)$ by the usual conserved-projection bound [4]. Portmanteau on the closed diagonal gives the opposite limiting upper bound W_0. Thus that finite-system weight also tends to W_0. The subtraction in (23) exhausts the limiting static weight for this observable and this state.

D Frequency moments and the resulting time dependence

Let $M_{2k} = \int \omega^{2k} \mathrm{d}\mu_O(\omega)$. In particular, $M_4 = \mathbb{E}\,\mathrm{tr}[(\mathrm{ad}_H)^2 O]^\dagger [(\mathrm{ad}_H)^2 O]$. The new moments are

$$M_2 = \frac{8\mathcal{J}_s^6}{\lambda^2}(1-q)(1+q+4\lambda q), \tag{26}$$

$$M_4 = \frac{8\mathcal{J}_s^8}{\lambda^3}(1-q)^2(5+6q+3q^2)(1+q+4\lambda q), \tag{27}$$

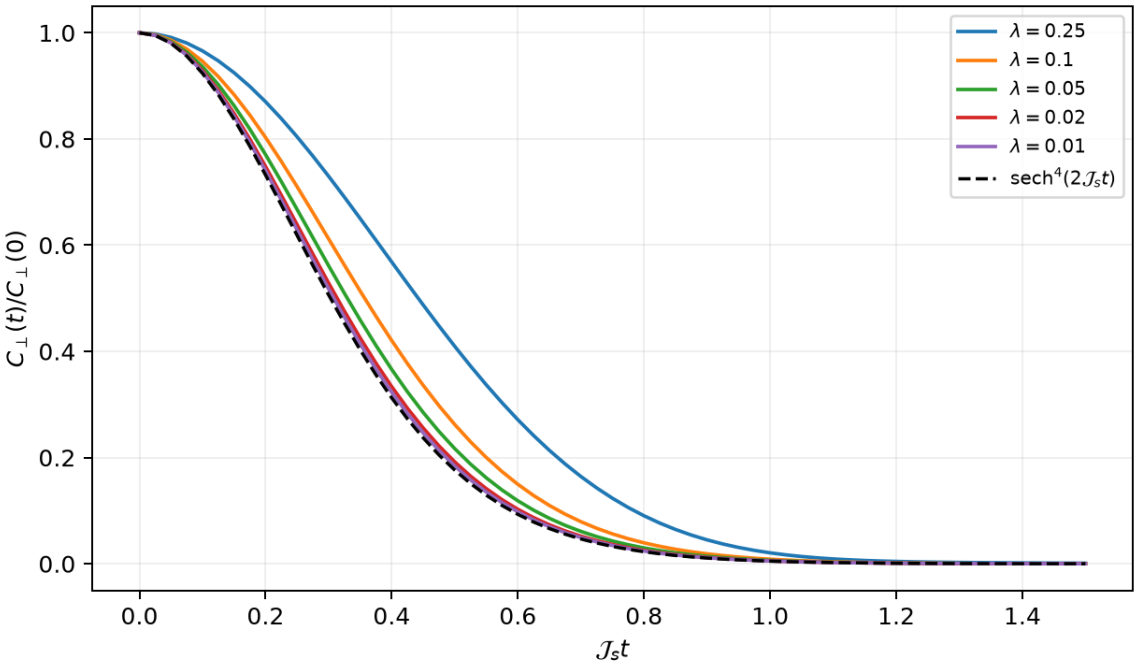
$$M_6 = \frac{8\mathcal{J}_s^{10}}{\lambda^4}(1-q)^3 R_5(q)(1+q+4\lambda q), \tag{28}$$

$$R_5(q) = 35+91q+115q^2+85q^3+40q^4+10q^5. \tag{29}$$

The direct marked expansion retains 390 fourth-moment configurations and 5250 sixth-moment configurations. In particular it does not discard the possible lower Gaussian components of $(\mathrm{ad}_H)^2 O$. It agrees with the independent-probe kernel moments multiplied by W_d. The small-λ normalized values are

$$M_2/V_0 \to 16\mathcal{J}_s^2, \qquad M_4/V_0 \to 896\mathcal{J}_s^4, \qquad M_6/V_0 \to 96256\mathcal{J}_s^6. \tag{30}$$

They agree with (10); the limiting fourth-to-second moment ratio $M_4 V_0/M_2^2$ is 7/2.



The plotted quantity is the normalized, conserved-subtracted correlator G_{q^2}, not the uncentered A correlator. All solid curves use the finite-λ kernel. The dashed curve is the known leading probe asymptotic. At $\lambda = .25, .1, .05, .02, .01$, the static fractions W_0/V_0 are respectively approximately $0.05442, 0.005576, 0.0008280, 0.00005920, 0.000007690$.

E Independent checks and numerical control

Exact checks. The saved symbolic calculation tests all 16 pairs (a, b) with $a + b = 0, 2, 4, 6$. It checks 5672 local graph rules and both identities (19), (20), as exact polynomials in q. The one-mark relation is checked through H^8. The eight-slot determinant was independently compared with CAR matrices in 1080 selected words, totaling 4320 rational coefficient comparisons. The supplementary set contains 262 nonzero mixed derivatives, including 23 negative values. The initially sampled set had only two nonzero values; the supplementary selection was added to address that weakness.

Direct Gaussian covariance sums, without using the chord reduction, give the following finite-N projection data in $w = 1$ units:

(N, p)	(O, O)	(O, B)	(B, B)	$(O, B)^2/(B, B)$
$(3, 1)$	12	15/2	15	15/4
$(4, 2)$	483/4	255/4	351/8	7225/78
$(5, 2)$	2015/4	2165/8	1765/8	937445/2824

The finite-N bound need not saturate. In particular $p = 1$ is conserved; it is not the fixed-positive-λ limit being taken here.

Kernel evaluation. In an orthonormal basis the Jacobi off-diagonal is $T_{n,n+1} = \sigma\sqrt{[n+1]_q}$, with $[n]_q = (1 - q^n)/(1 - q)$. We evaluate

$$G_{q^2}(t) = \sum_{n\geq 0} q^{2n} |\langle n|e^{-itT}|0\rangle|^2. \tag{31}$$

For a truncation to L states and $a = E_{\max}|t| < L + 1$, agreement of the first L Taylor coefficients gives the exact-arithmetic bound

$$|G(t) - G_L(t)| \leq 4 \sum_{k=L}^{\infty} \frac{a^k}{k!} \leq \frac{4a^L}{L!} \frac{1}{1 - a/(L+1)}. \tag{32}$$

For $\mathcal{J}_s|t| \leq 1.5$, the selected sizes $L = 40, 67, 109, 233, 437$ make this bound smaller than 10^{-12}. Repeating with $L + 16$ changes the values by at most 4.2×10^{-15}.

Separately, 96- and 192-node Gauss–Legendre integrations of (8), using infinite-product tail scales below 10^{-16}, test $\lambda = .25, .1$. Agreement with Jacobi evolution is better than 3.5×10^{-14} at the recorded times. Moments through order six have relative residual below 3.5×10^{-14}. These floating-point residuals are empirical; the analytic truncation bound does not certify eigensolver rounding or quadrature error.

F Scope, provenance, and continuation

The result is for the full zero-chemical-potential infinite-temperature trace, the full complex Gaussian ensemble, and the stated normalization. The proof takes double scaling at each fixed moment order and then uses positivity and compact moment determinacy. It proves a two-point spectral reduction, not equality of the complete operator algebras or of higher correlators with an independent probe. Finite-N recurrences and time scales growing with N are not covered by fixed-time weak convergence. The small-λ expression is a subsequent limit of the known probe kernel, not a uniform finite-N estimate.

The conserved projection and q-Mehler kernel are established tools. The calculation specific to this note is the marked endpoint exchange reduction and its application to A. No exhaustive priority review has been performed. The finite-word evidence supports the microscopic insertion and signs; the all-order identity is supplied by the proof, not inferred from the finite tests.

The immediate physics task is to compare the $\Delta = 2$ kernel with the Rindler observable proposed in the Susskind programme, fixing string versus cosmic time, smearing, and state. The next microscopic task is the three- and four-insertion singlet problem: shared Hamiltonian couplings can generate connected terms that the two-point reduction alone does not determine.

Reproducibility. Source scripts and exact/numerical outputs are in the project's `code` and `results/susskind-singlet-kernel` directories. The principal scripts are:
`singlet_spectral_graphs.py`,
`check_singlet_spectral_kernel.py`,
`check_singlet_kernel_nonzero.py`,
`run_singlet_kernel.py`. Input and output hashes, compiler logs, and page-review records accompany the local stage. No Git commit or push accompanies this note.

References

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